

The Theory of Permyakov Codes

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Abstract

The Permyakov Code is a novel universal prefix code designed for extremely efficient integer compression. It serves as an alternative to classical bit-level codes like Elias Omega and byte-level encodings like Varint (LEB128 / VByte). Its primary theoretical breakthrough is substituting the recursive bit-length encoding ($\log_2$) found in Elias Omega with a recursive Hamming weight calculation (**popcount**), and applying an internal block reversal to elegantly solve the inherent "trailing zeros" decoding ambiguity.

1 Mathematical Formulation

For any integer $N > 0$, Elias Omega recursively encodes the bit-length sequence: $L_0 = \lfloor \log_2(N) \rfloor$, $L_1 = \lfloor \log_2(L_0) \rfloor$, and so on.

The Permyakov Code replaces this with a modified popcount sequence. Since the Most Significant Bit (MSB) of any positive integer is always 1, and because Permyakov blocks are written and read in reverse (from LSB to MSB), the decoder can naturally infer the final 1 bit of every block. Thus, the effective "weight" to be encoded collapses faster: $W_0 = \text{popcount}(N) - 1$, $W_1 = \text{popcount}(W_0) - 1$, and so on.

1.1 The Theorem of Strict Dominance

The fundamental property guaranteeing the superior compression of Permyakov Code over Elias Omega is the following strict inequality, which holds for all $N > 1$:

$$\text{popcount}(N) - 1 < \lfloor \log_2 N \rfloor \quad (1)$$

In the absolute worst-case scenario where an integer consists entirely of set bits ($N = 2^k - 1$), the modified popcount exactly matches the Elias Omega bit-length:

$$\text{popcount}(2^k - 1) - 1 = k - 1 = \lfloor \log_2(2^k - 1) \rfloor \quad (2)$$

In all other scenarios, the Permyakov chain converges strictly faster. As a result, **Permyakov Code strictly dominates Elias Omega**: its structural overhead is always less than or equal to Omega across the entire integer domain.

2 Visualizing the Overhead

We can observe this theoretical dominance by plotting the structural metadata overhead (total encoded bits minus the raw binary length of the integer) for both algorithms up to 2^{60} .

2.1 Best Case Scenario ($N = 2^k$)

When a number is a power of 2, it contains exactly one 1 bit. Consequently, its popcount is 1. The Permyakov Code overhead collapses instantly into its absolute minimum of 3 bits (a 2-bit prefix 01 and a 1-bit terminator 1).

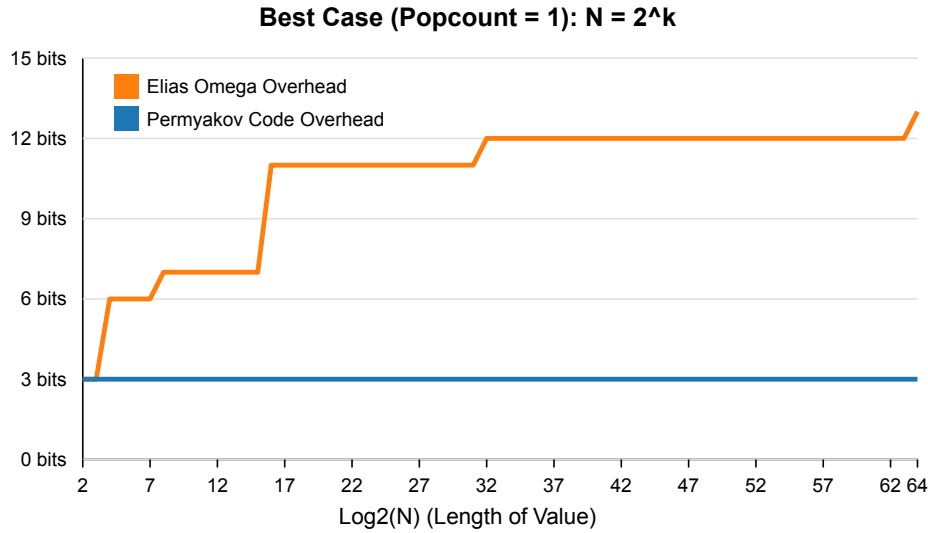


Figure 1: Best Case: $N = 2^k$. While Elias Omega overhead grows logarithmically, Permyakov Code overhead remains perfectly constant at 3 bits.

2.2 Worst Case Scenario ($N = 2^k - 1$)

In the absolute worst-case sequence, all bits in the raw binary representation are set to 1. This forces the popcount to be equal to the bit-length.

3 Conclusion

The Permyakov Code provides mathematical optimality for prefix integer coding. By replacing $O(\log \log N)$ bit-length convergence with hardware-accelerated **popcount** convergence minus 1, it achieves significantly higher compression density with zero edge-case losses against Elias Omega.

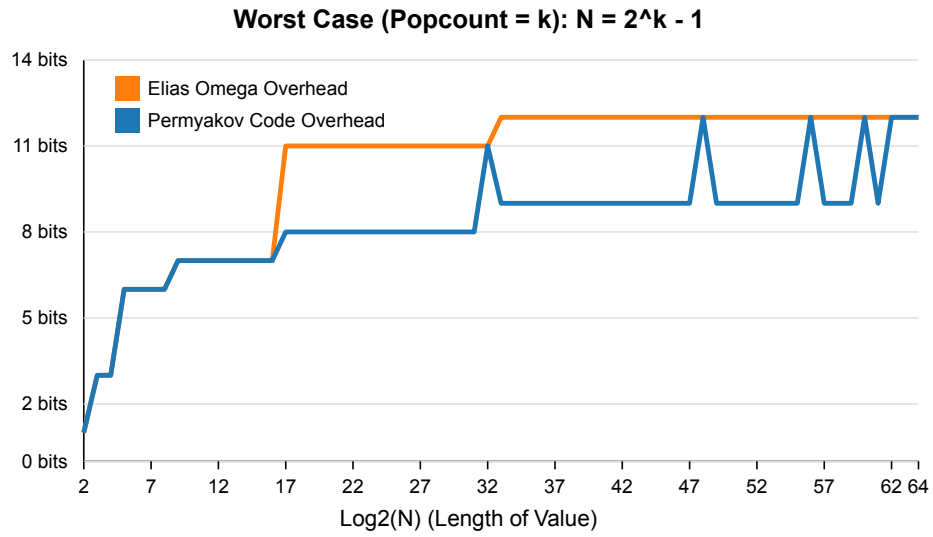


Figure 2: Worst Case: $N = 2^k - 1$. Even in its mathematically proven worst case, the Permyakov Code line flawlessly aligns with or drops below the Elias Omega line, confirming strict dominance.